Justin Chang

Electrical Engineering Student

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Objective

I am a fourth year Electrical Engineering student specializing in Semiconductors and Optoelectronics with hands-on experience in hardware development of a smart plug energy monitor and lots of academic experience in electrical knowledge. I served as hardware lead and was able to successfully deliver a working final product in the given time and budget constraints. During that process, I have designed, tested, troubleshooted, built the hardware side of our product, including using C++ to code a microcontroller and designing a PCB board. As the hardware lead, I also took responsibility for documentation and quick communication to keep everyone updated. During my time at UCI, I also got a lot of experience in the lab, using oscilloscopes, multimeters, and signal generators to build and analyze circuits. I hope to apply and hone these skills in the industry.

Education

Irvine Valley College - Irvine, CA

September 2022 - June 2024

- Completion of Lower Division Coursework
- GPA: 3.9

University of California, Irvine - Irvine, CA

September 2024 - Present

- Relevant Coursework: Introduction to Digital Systems, Introduction to Digital Logic, Network Analysis II, Engineering probability, Continuous-Time Signals and Systems, Electronics I, Electronics II, Electronics III, Electronics I Lab, Electronics II Lab, Electronics III Lab, Fundamentals of Solid-State Electronics and Materials, Introduction to Control Systems and Lab, Principal Materials Science, Fundamentals of Optics, Power Systems, Communications in the Professional World, Drones
- GPA: 3.4

Projects

Senior Design Project - Smart Plug Energy Monitor (ZotPlug) - Hardware Lead - 5-person team

September 2025 - March 2026

- Led hardware development of a smart plug monitor using an ESP32 and metering IC to measure the user's real-time voltage, current, and power consumption
- Finished with a fully end to end system with a validated/working perfboard design able to communicate and controlled through our app and website with PCB board designed and made for potential iterations.
- Collaborating as a 5-person group on Visual Studio code and GitHub for version updates
- Our design consists of an ESP32 and metering IC for accurate current, voltage, and power readings for the user at both high and low currents in order to be able to implement our software to turn on and off depending on previous readings.
- Many projects before us used a clamp current sensor (SCT-013-050) and ESP32 directly, which has large errors (15% even at high currents), while our metering IC and ESP32 set up has reduced it to around 3% for both current and power.

Work Experience

Math Teacher Assistant/Grader

June 2023 - September 2023

- Assisted an assigned instructor to help and grade and answer questions on homework, projects, and exams of over 200+ students.
- Good communication with other graders for fair grading criteria for all students was necessary.
- Communicating and solving issues with students on a timely basis was also very important.

Skills

- Programming Languages: C++
- Related Skills: Visual Studio Code, MatLab, GitHub, worked with ESP32 and Raspberry Pico Pi W, EasyEDA, Signal Generators, Electronic Test Equipment, Microsoft Office, AutoCAD, Hands-on Lab experience, Digital Oscilloscope, Digital Multimeter, DC Power Supplies, AC Power Supplies, Test Lab Experience, Cable/Wiring Schematics, Analog Circuit Analysis, PCB schematic and building, Soldering Experience, Experience with 120 volts